A PROJECT PROPOSAL REPORT ON

careerNexus:
CONNECTING TALENT WITH OPPORTUNITY



In Partial Fulfillment of Requirements for Bachelor of Science in Computer Science and Information Technology

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1. INTRODUCTION

One of the most urgent problems in the employment ecosystem has long been finding the proper job or hiring the best applicant. Employers find it challenging to find applicants who actually fit their job requirements, while job seekers frequently struggle to find positions that match their experience, abilities, and career goals. Conventional job portals have mostly stayed the same and have an antiquated methodology; they frequently lack the sophistication required to offer intelligent job matching, tailored recommendations for jobs or candidates, and efficient user contact. Due to this gap, many chances are lost, job postings are irrelevant or of poor quality, and the hiring process is time-consuming and ineffective for all parties.

careerNexus is a cutting-edge, AI-powered employment portal created especially to close these gaps in the hiring landscape in response to these constraints and the growing need for more intelligent, tech-driven solutions. careerNexus provides a smooth and user-friendly experience by utilizing the capabilities of the MERN stack, which includes Express.js and Node.js for scalable backend operations, React.js for a dynamic and responsive user interface, and MongoDB for effective data storage.

The platform serves the specific requirements of three main user groups: employers, who can post job openings and find the best candidates through intelligent filtering and recommendations; job seekers, who can investigate relevant job opportunities and get AI-driven recommendations based on their profiles; and administrators, who supervise and manage the entire system to guarantee seamless operation and compliance. These characteristics work together to make careerNexus a potent tool that aims to revolutionize the employment process by bringing efficiency, customisation, and automation at every stage.

2. PROBLEM STATEMENT

Because traditional job portals lack cognitive, context-aware technologies, resumes and job listings frequently do not match well. Candidates commonly obtain generic or irrelevant job recommendations that do not match their talents, experiences, or career aspirations because these platforms usually offer little to no customisation. In addition to making users frustrated, this drastically lowers the likelihood of finding fulfilling job matches. From the employer's point of view, the position is just as difficult. It is quite challenging for recruiters to sort through a huge pool of applications and find those who are truly qualified for the position because many traditional platforms have inflexible and limited filtering and search features.

Furthermore, the majority of conventional job portals use simple keyword-based search algorithms that are unable to comprehend the semantics or context of a job description or a candidate's expertise. For instance, due to keyword mismatches, a candidate with project management experience at a tech company would not be suggested for a project management position in a different industry. High-quality matches between companies and candidates are hindered by this lack of semantic understanding, which also leads to ineffective job discovery.

Moreover, when it comes to resume screening and analysis, these portals usually lack any automation or intelligence. For recruiters, manually assessing resumes according to experience and skill sets becomes a laborious and time-consuming procedure, particularly when hundreds or thousands of applications are received. These antiquated methods are no longer enough in the fast-paced, technologically advanced world of today.

careerNexus offers a game-changing solution in this situation. Semantic matching between resumes and job descriptions is made possible by the platform's robust cosine similarity-based AI technology. This approach examines the true meaning and context of words and phrases rather than relying just on surface-level keyword matches. As a consequence, companies are shown the most qualified applicants based on a

sophisticated review of their resumes, and job seekers are matched with opportunities that actually fit their skills. The entire effectiveness of the hiring process is increased by this clever matching procedure, which also guarantees a far better platform experience for businesses and job seekers.

3.OBJECTIVES

- To create a fully functional employment portal with role-based access and management that offers different dashboards and rights to administrators, businesses, and job seekers.
- To apply cosine similarity-based AI-based semantic matching to match resumes with job descriptions while guaranteeing recommendations that are highly relevant.
- To provide individualized employment recommendations by examining user profiles, preferences, experiences, and talents.
- To make sophisticated search and filtering tools available so that companies can quickly identify the best applicants from a huge applicant pool.
- To minimize manual hiring effort by automating resume analysis and screening according to candidate experience, abilities, and credentials.
- To utilize React.js to develop a responsive, user-friendly interface that is accessible to people with different levels of technical expertise and optimized for all devices.

4. METHODOLOGY

4.1 LITERATURE REVIEW

The increasing volume of job applications has made resume screening a challenging and time-consuming task for HR departments[1]. Traditional manual methods are often biased and inefficient, leading to the potential overlooking of highly qualified candidates [2]. To address this, many researchers have explored AI and NLP-based solutions for automating the screening process [3]. Ransing et al. [4] proposed a stacked model that combines rule-based and machine learning approaches to improve resume ranking accuracy while minimizing bias. Similarly, Bhardwaj et al. [5] used LSTM networks to capture the sequential nature of resume content, achieving 94% accuracy, although at the cost of high computational power. Pant et al. [6] demonstrated how combining NLP with word matching and character positioning can effectively extract key features from resumes, though the approach struggles with false positives and nuanced skills. Pradeep et al. [7] developed a content-based recommendation system using k-NN and NLP, showing high accuracy but limited file format support and potential algorithmic bias. Mahdi et al. [8] further enhanced resume matching by applying BERT-based keyword extraction from job descriptions, improving semantic understanding. Considering these advancements, careerNexus adopts a cosine similarity-based algorithm to semantically match resumes and job descriptions efficiently and objectively [3].

4.2 REQUIREMENT ANALYSIS

4.2.1 FUNCTIONAL REQUIREMENTS

- The system must allow **job seekers, employers, and administrators** to securely register and log in using valid credentials.
- Job seekers must be able to **create and manage their profiles**, upload resumes, and update their skills, experiences, and preferences.

- Employers must be able to **post job listings**, update job details, and manage applications from interested candidates.
- The system must utilize a **cosine similarity algorithm** to semantically match job listings with resumes, enhancing job-candidate relevance.
- Job seekers must receive **personalized job recommendations** based on their profiles, skills, and past interactions.
- Employers must have access to a **dashboard** that allows them to view applicant details, filter candidates, and shortlist based on relevance.
- Administrators must be able to **manage users, moderate content**, and oversee the system to ensure fair and secure operations.
- The system must include **advanced search and filtering options** for both employers and job seekers (e.g., by location, experience level, salary, job type).
- Users should be able to **save jobs**, track application status, and receive alerts for new job postings matching their interests.

4.2.2 NON-FUNCTIONAL REQUIREMENTS

- The system must ensure **real-time job matching and search operations** with a response time of under **3ms**, enabling smooth user interaction.

- The platform should offer **dark mode and light mode toggling** to enhance user experience and accessibility.
- The platform should support **24x7 availability**, ensuring that users can access services at any time, even though peak hours are during business times.
- The system should be **fully responsive and device-compatible**, functioning seamlessly across desktops, tablets, and smartphones.
- The system must comply with **data protection regulations** and digital privacy standards applicable to user data handling and digital transactions.

4.3 FEASIBILITY STUDY

4.3.1 ECONOMIC FEASIBILITY

• **Software Development**: Free Third-Party services are used for software development ensuring no cost to the development team.

• **Hardware Requirements**: The application is hosted on free hosting platforms ensuring zero investment from developers involved.

• **Licenses and Tools**: Payment gateways, development environments are accessed with educational intentions helping in ensuring no cost to the group members.

4.3.2 TECHNICAL FEASIBILITY

Technology Requirements

- **Software :**
    - ➢ Programming Languages: HTML, CSS, JavaScript are available easily.
    - ➢ Framework and Libraries: ReactJS, ExpressJS, NodeJS are also open source and free.
- **Database :** MongoDB is a free to use NoSQL Database system.
- **Security :** SSL certificates for secure connections, MFA integration for user authentication can easily be integrated without hassle.

Technical Skills

4.3.3 OPERATIONAL FEASIBILITY

- The system must be designed with separate dashboards for **Job Seekers**, **Employers**, and **Admins**, ensuring intuitive navigation tailored to each user's needs.
- Role-based permissions should ensure that job seekers, employers, and admins **only access features relevant to their role**, enhancing both usability and security.
- The use of **AI-based cosine similarity** should ensure that resume-job matches happen efficiently, providing **real-time recommendations** to users.

- The system must be scalable and maintainable as the number of users increases.
- The system must be responsive in order to be compatible on all screen sizes (PC, tablets, mobile phones,etc).

4.4 SYSTEM DESIGN

4.4.1 DEVELOPMENT METHODOLOGY

The Agile methodology will be applied throughout the project to facilitate flexibility and responsiveness to change. Agile focuses on iterative development, where the project is broken down into smaller, manageable units of work called sprints, typically lasting one to two weeks. Each sprint will result in a functional piece of the application that can be tested, reviewed, and refined based on user feedback.

This iterative process allows for continuous improvements and ensures that the development stays aligned with user needs and project goals. Regular sprint reviews and retrospectives will provide opportunities for the team to assess progress, identify any challenges, and make necessary adjustments. Agile's emphasis on collaboration and communication among team members and stakeholders ensures that the project remains on track and any issues are addressed promptly.

In this project, Agile will also enable the rapid incorporation of changes, particularly as the development involves integrating complex features like real-time messaging and encryption. By following Agile, the project team can ensure that the final product meets user expectations while delivering high-quality software that is both functional and secure.

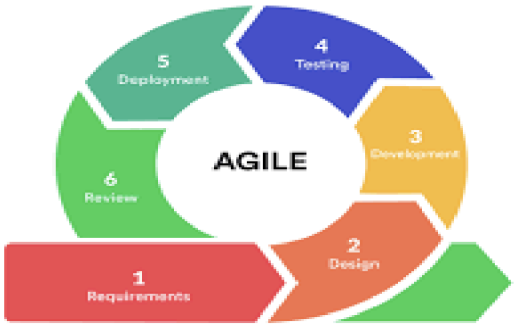


Fig: Agile Methodology

4.4.2 TECHNOLOGY

Frontend: The front-end of the application will be built using React.js for efficient UI rendering, Redux Toolkit for state management, and Tailwind CSS for responsive and modern styling.

Backend: Backend of careerNexus will utilize NodeJS and ExpressJS, with MongoDB as the database.

4.4.3 ALGORITHM

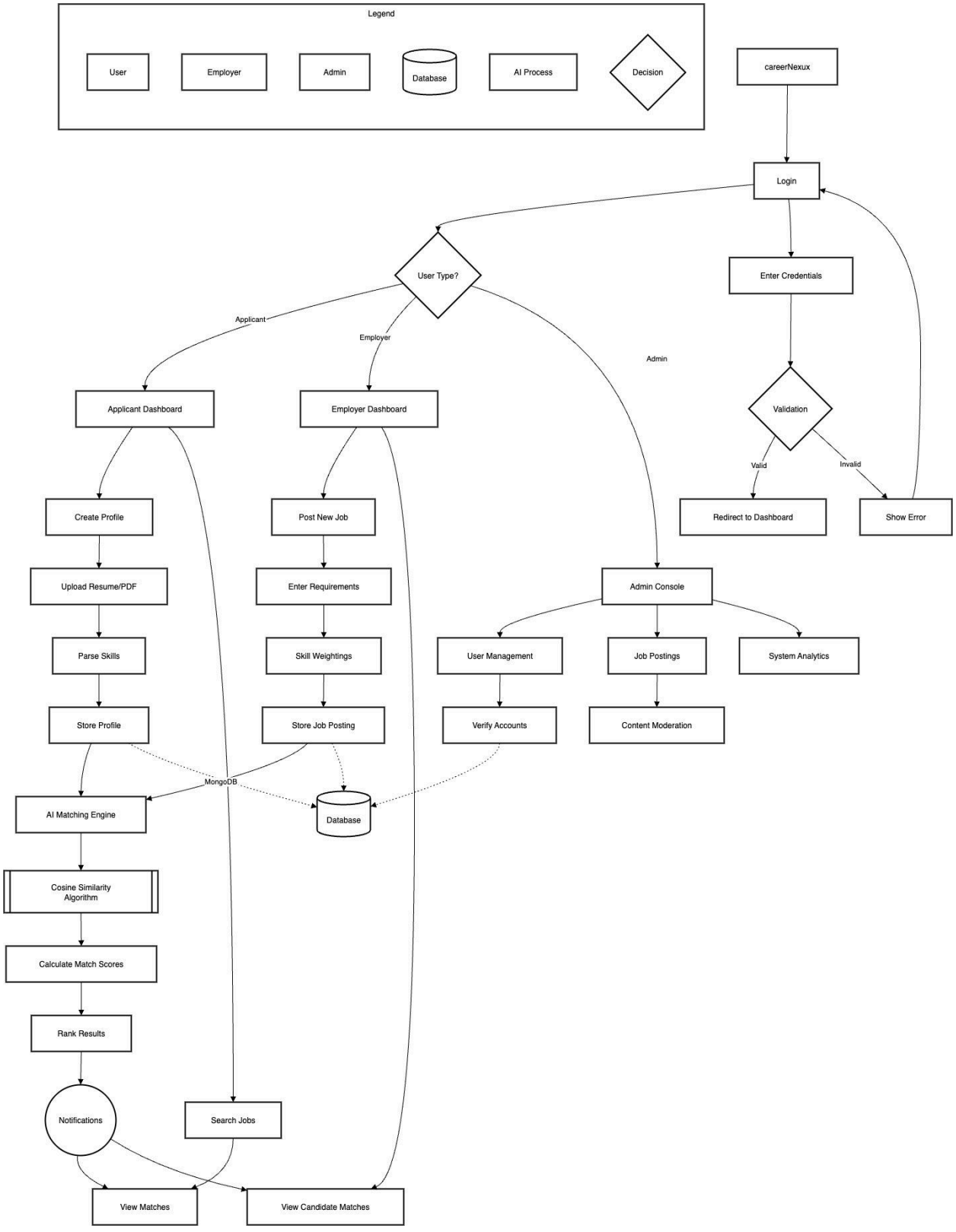
The **cosine similarity algorithm** measures the similarity between two non-zero vectors by calculating the cosine of the angle between them. We use this algorithm to convert the job and the resume into a vector of terms (words/phrases). Then, using the formula, we calculate the angle between these two vectors, which might range from -1 to 1(-1≤ cosθ ≤1).

Mathematically,

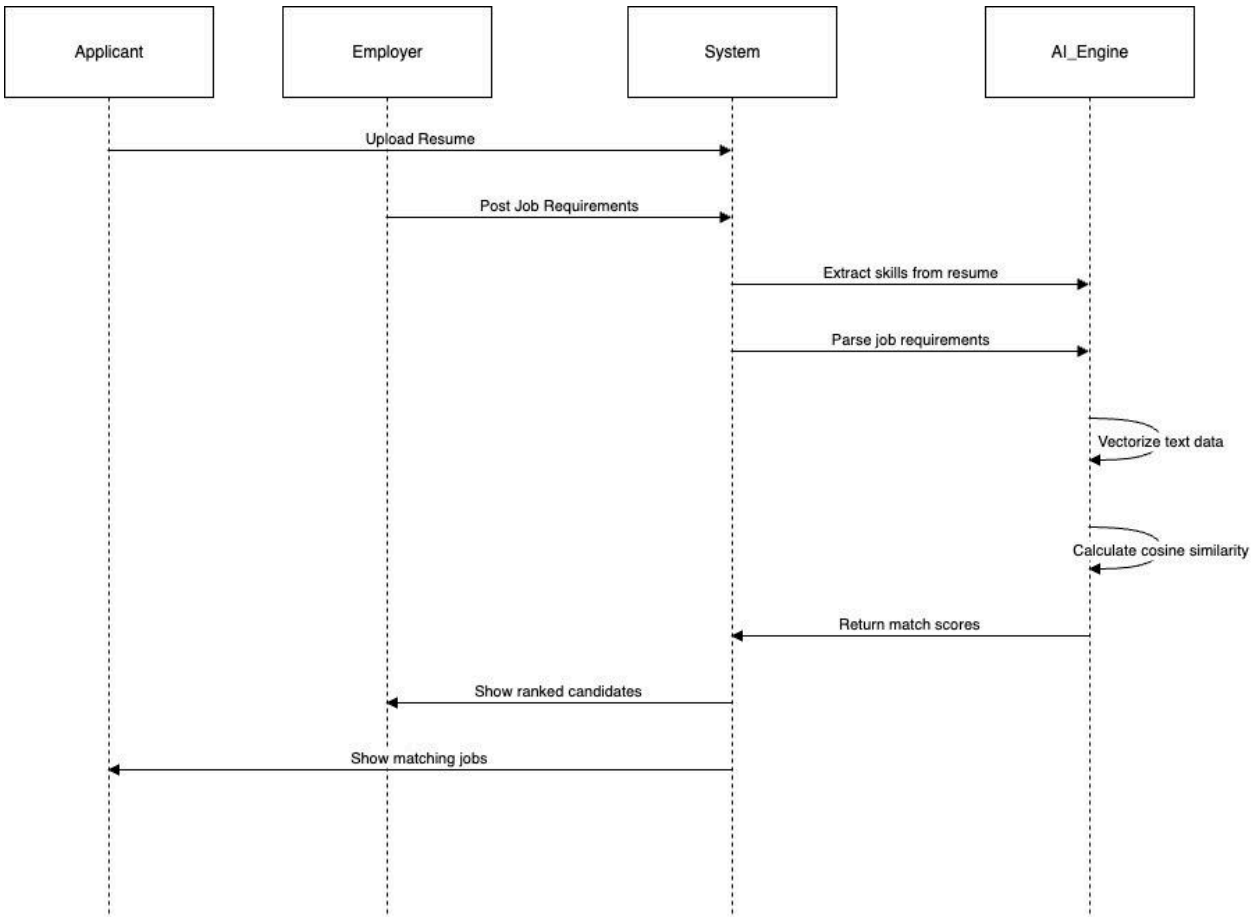
$$S_c(x,y) = \frac{x.y}{|x||y|}$$

where,
x.y = product (dot) of the vectors 'x' and 'y'.
|x|= magnitude of vector x.
|y|= magnitude of vector y.

4.4.4 FLOWCHART



4.4.5 SEQUENCE DIAGRAM



5. EXPECTED OUTPUT

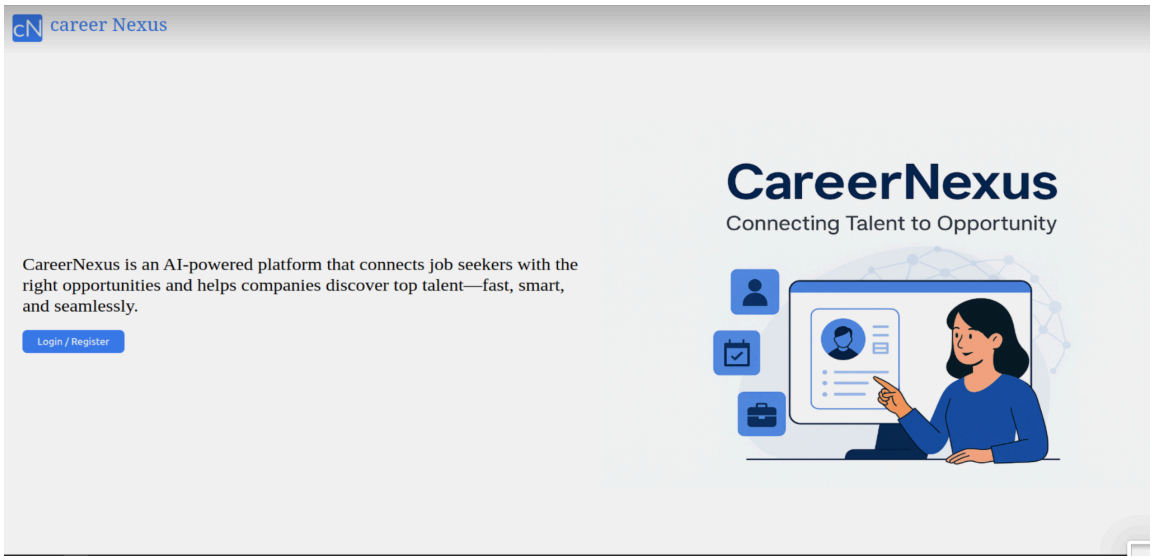


Fig: Landing Page

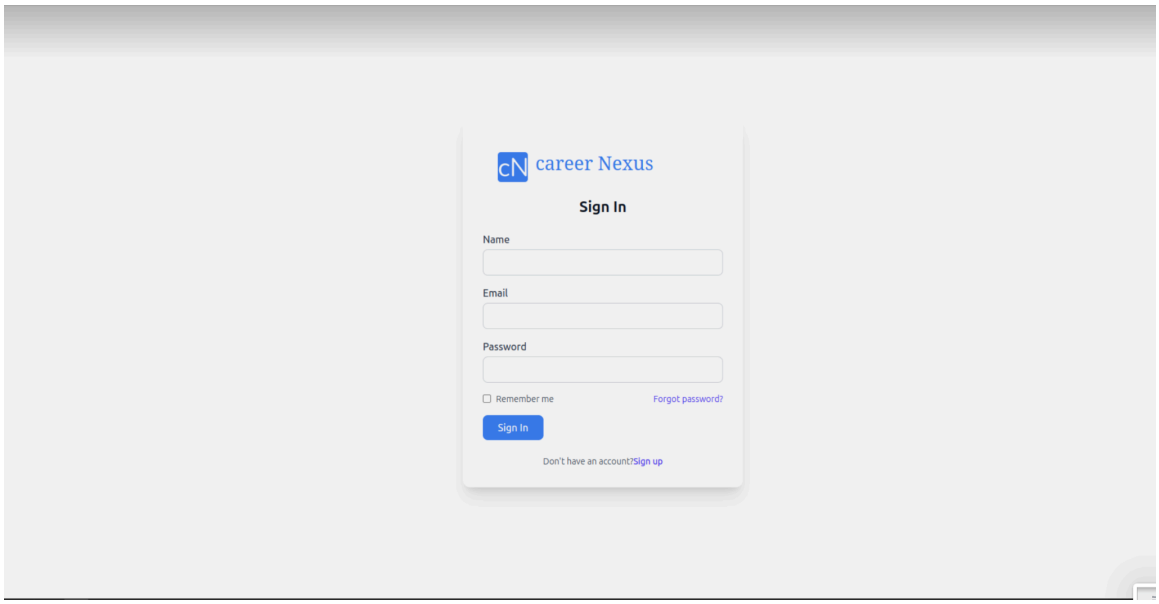


Fig : Login Page

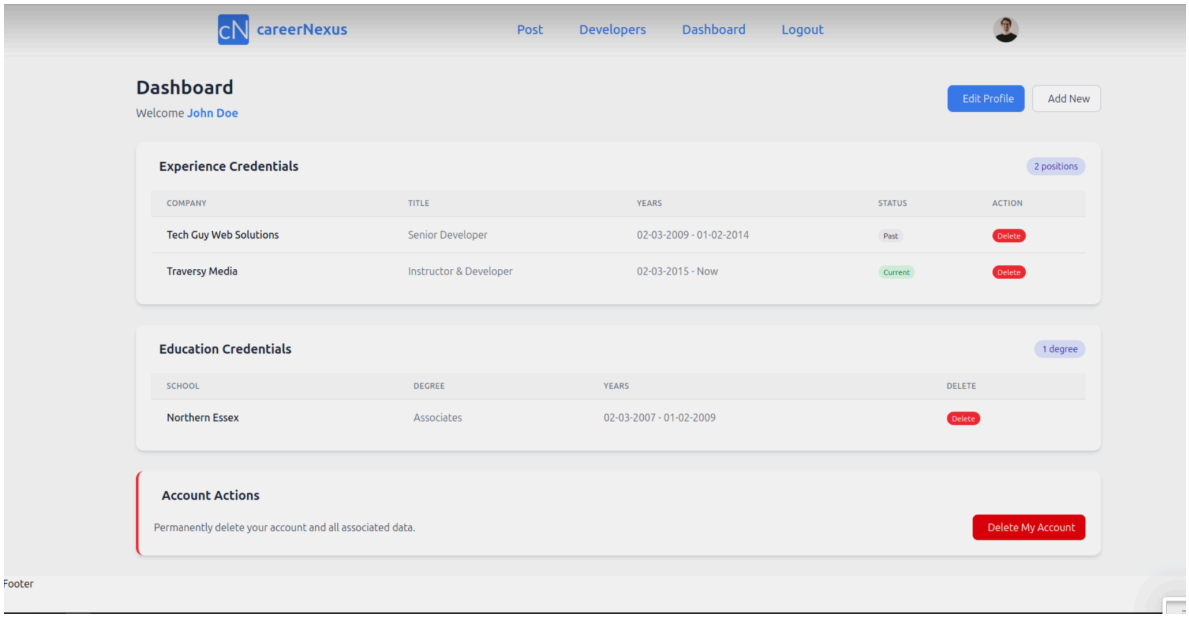


Fig : Admin Dashboard

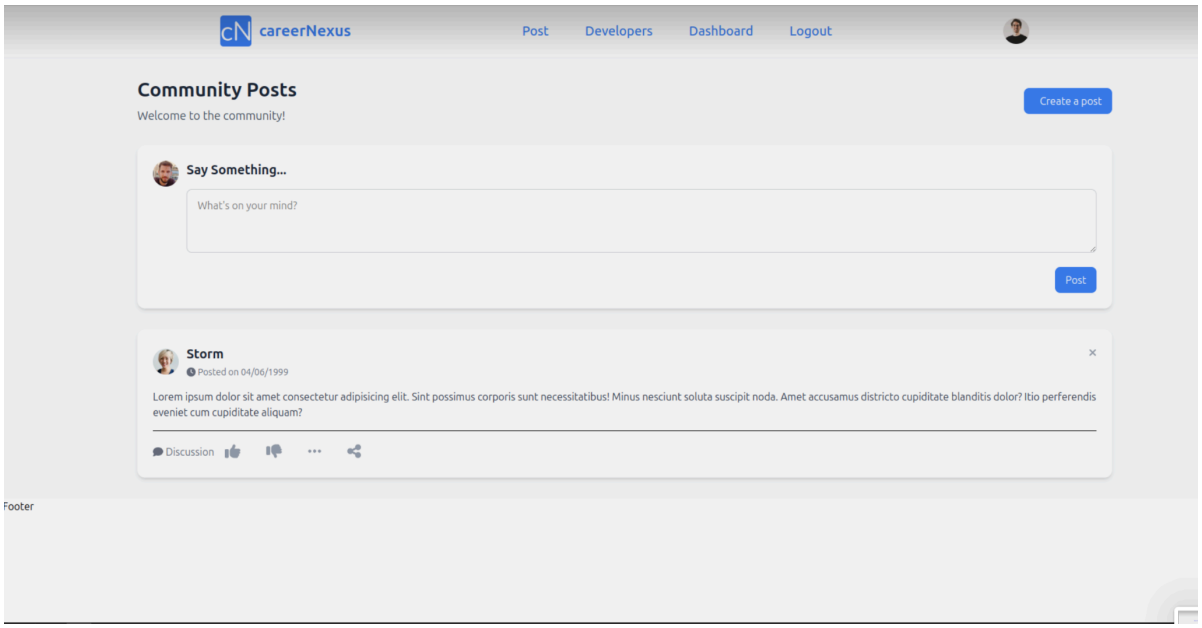


Fig: Community Post

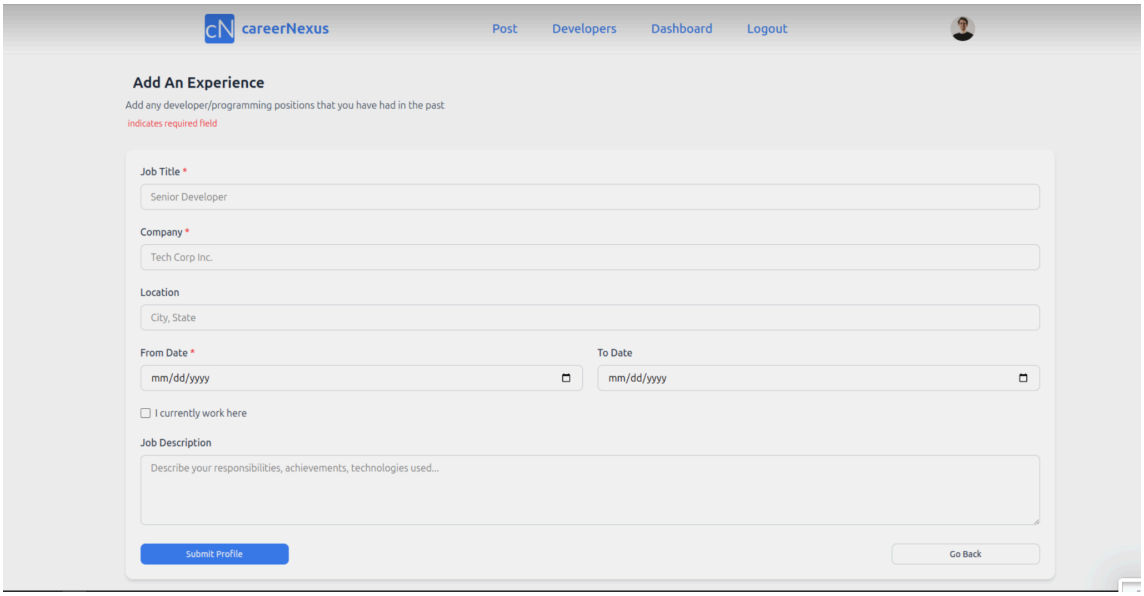


Fig: Employee adding their experience

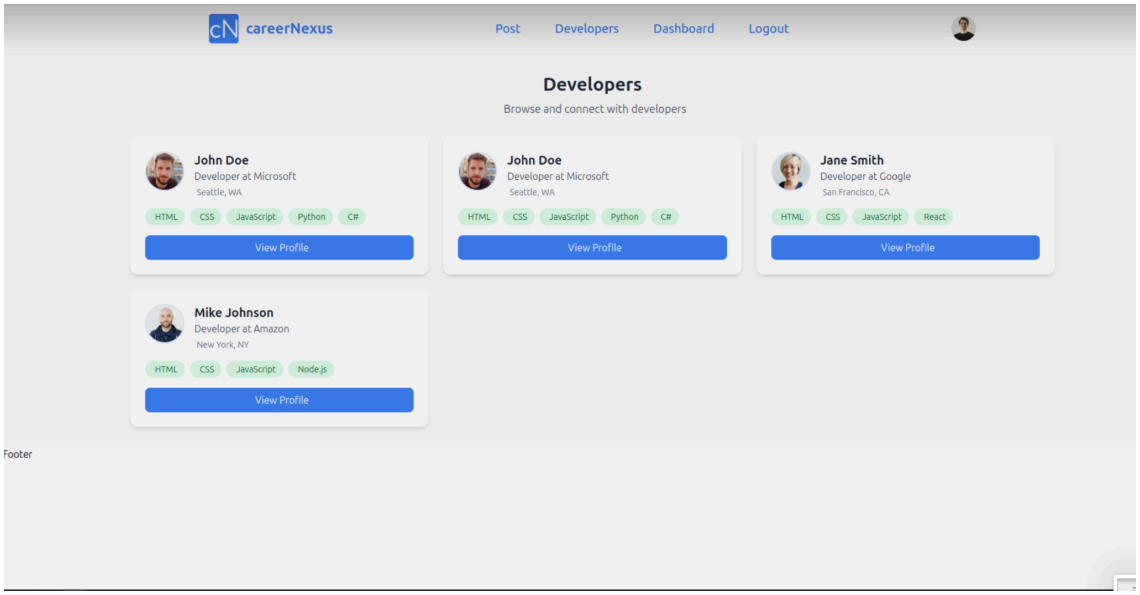


Fig: List of matched employees

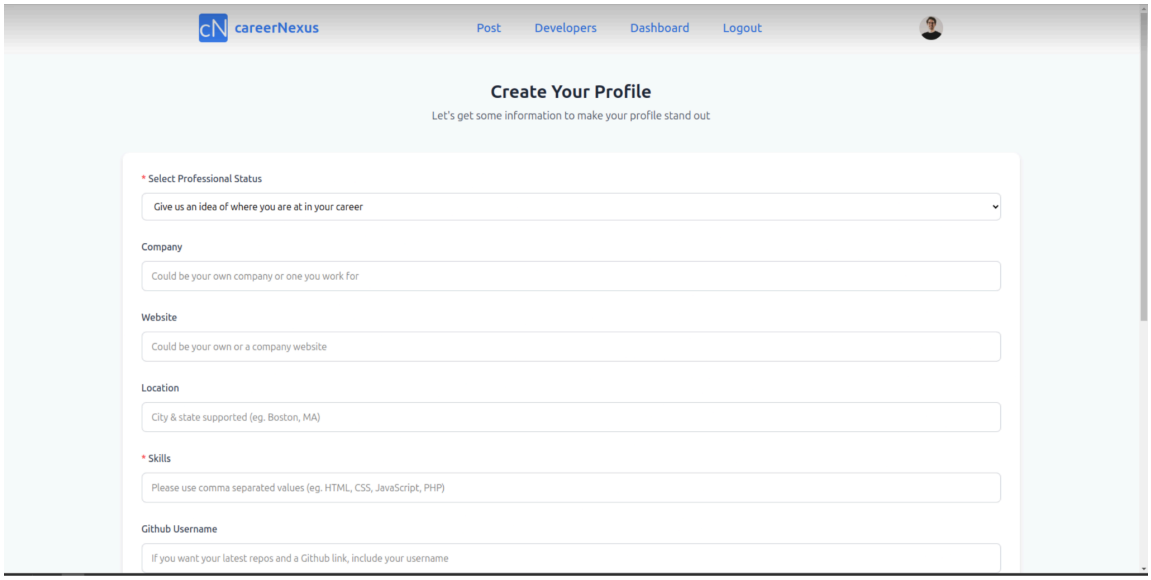


Fig: Creating a profile

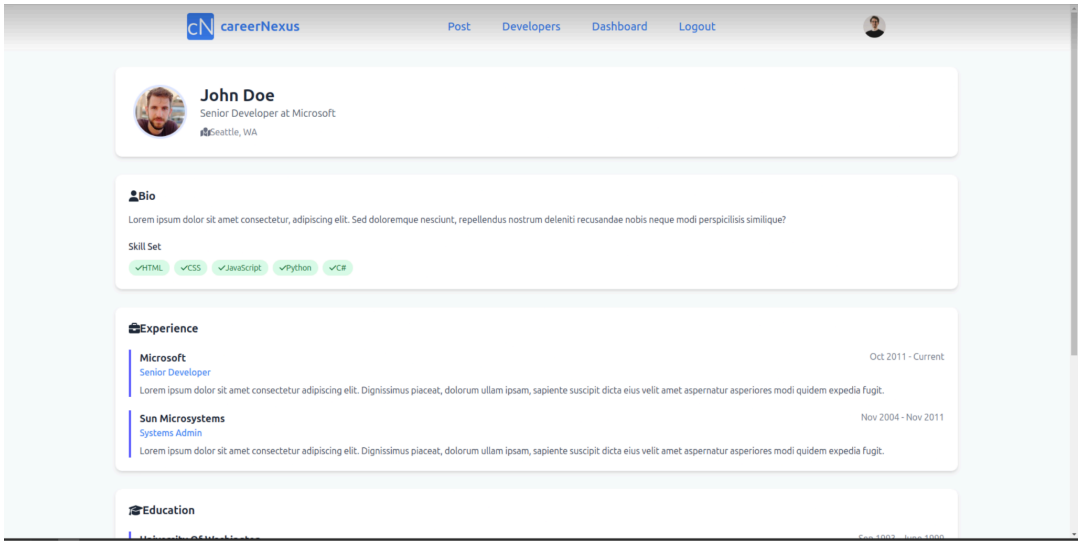


Fig: Employee Profile

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